Efficacy of Mobocertinib and Amivantamab in Patients With Advanced Non–Small Cell Lung Cancer With *EGFR* Exon 20 Insertions Previously Treated With Platinum-Based Chemotherapy: An Indirect Treatment Comparison

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Abstract

Mobocertinib and amivantamab are used to treat *EGFR* ex20ins+ non-small-cell lung cancer (NSCLC) but have not been compared in a head-to-head trial. We conducted a matching-adjusted indirect comparison to compare their efficacy. Efficacy outcomes were generally comparable for mobocertinib and amivantamab. These findings confirm the efficacy of different treatment options for patients with *EGFR* ex20ins+ NSCLC.

Introduction: Exon 20 insertions (ex20ins) mutations of the *EGFR* gene account for 1% to 2% of all non–small-cell lung cancers (NSCLCs). Targeted therapies have been developed to treat this cancer type but have not been studied in head-to-head trials. Our objective was to use a matching-adjusted indirect comparison (MAIC) to assess the efficacy of mobocertinib and amivantamab in patients with NSCLC *EGFR* ex20ins mutations who were previously treated with platinum-based chemotherapy. **Materials and Methods:** An unanchored MAIC was conducted to estimate the treatment effects of mobocertinib and amivantamab using individual-level data from the mobocertinib phase I/II single-arm trial (NCT02716116) and published data from the amivantamab single-arm CHRYSALIS trial (NCT02609776). Confirmed overall response rate (cORR), progression-free survival (PFS), overall survival (OS), and duration of response (DoR) were assessed. **Results:** Both trials were comparable in terms of study population, study design, and outcome definitions and included 114 patients who received mobocertinib and 114 patients who received amivantamab. After MAIC weighting, all reported baseline characteristics were balanced between mobocertinib and amivantamab. The weighted odds ratio (OR) [95% confidence interval (CI)] comparing mobocertinib to amivantamab was 0.56 (0.30-1.04) for independent review committee (IRC)-assessed cORR and 0.98 (0.53-1.82) for investigator (INV)-assessed cORR. The weighted hazard ratio (HR) comparing mobocertinib to amivantamab was 0.74 (0.51-1.07) for IRC-assessed PFS, 0.92 (0.57-1.48) for OS, and 0.59 (0.30-1.18) for INV-assessed DoR. **Conclusion:** MAIC analysis showed that mobocer-

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tinib and amivantamab had similar efficacy in patients with NSCLC harboring *EGFR* ex20ins mutations whose disease progressed during or after platinum-based chemotherapy. These findings may benefit patients by supporting future treatment options.

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Introduction

Non-small cell lung cancer (NSCLC) is the most common form of lung cancer, accounting for approximately 85% of lung cancers diagnosed each year worldwide.[1] Exon 20 insertions (ex20ins), which cause constitutive activation of the epidermal growth factor receptor (EGFR) pathway, are the third most common *EGFR* genetic mutations in NSCLC and account for 1% to 2% of all NSCLC cases.[2] NSCLC with *EGFR* ex20ins mutations are less sensitive to conventional EGFR tyrosine kinase inhibitors (TKIs) than NSCLC with other more common *EGFR* mutations.[3-5] For NSCLC with *EGFR* ex20ins mutations, the most commonly used initial systemic therapy is platinum-based doublet chemotherapy,[6] but over the past decade newer and more effective targeted therapies have been developed.[2,6]

Recently, two *EGFR* ex20ins targeted therapies—mobocertinib, an oral EGFR TKI, and amivantamab, an intravenously administered monoclonal antibody targeting *EGFR* and c-Met[7]—received accelerated US FDA approval for the treatment of locally advanced or metastatic NSCLC that has progressed during or after platinum-based chemotherapy.[8,9] In a phase I/II clinical study (NCT02716116),[10] treatment with mobocertinib resulted in an investigator (INV)-assessed response rate of 35% (95% confidence interval [CI]: 26%-45%), an independent review committee (IRC)-assessed response rate of 28% (95% CI: 20%-37%), and had a manageable safety profile. Similarly, in the CHRYSALIS phase I clinical study,[11] amivantamab had a blinded independent central review-assessed response rate of 40% (95% CI: 29%-51%), an INV-assessed response rate of 36% (95% CI: 25%-47%) and showed a tolerable safety profile.

In the absence of head-to-head studies comparing mobocertinib and amivantamab, an indirect treatment comparison (ITC) of efficacy between these treatments can help guide treatment choices. Standard ITC methods such as Bucher indirect comparisons[12] and network meta-analyses[13] are not feasible because the mobocertinib and amivantamab trials are single-arm and lack a common comparator. In addition, individual-level patient data in the amivantamab trial is not available; therefore, matching-adjusted indirect comparison (MAIC) is the most appropriate choice to compare the efficacy of mobocertinib and amivantamab.[14]

In this study, we conducted an MAIC to compare the efficacy of mobocertinib and amivantamab in patients with NSCLC with *EGFR* ex20ins mutations who had been previously treated with platinum-based chemotherapy.

Materials and Methods

Study Design

An unanchored MAIC was conducted to estimate the relative treatment effects of mobocertinib vs. amivantamab using individual patient data from the mobocertinib phase I/II trial (NCT02716116) and the published aggregate level results from the amivantamab CHRYSALIS trial (NCT02609776).[10,11]

A direct comparison of treatments investigated in separate single-arm studies is likely to be biased by cross-study differences in patient populations on key prognostic factors and treatment-effect modifiers. When individual patient data are available for one population (index trial) and aggregate level data for another (the comparator trial), the MAIC can be used to estimate comparative efficacy of the two treatments.[14-17] The MAIC method weights patients in the index trial such that their weighted average characteristics match those of patients in the comparator trial.[14-17] After MAIC adjustment, relative efficacy can be estimated across balanced trial populations.

The clinical outcomes included confirmed overall response rate (cORR: the percentage of patients who have a partial or complete response to treatment), duration of response (DoR: the time from study onset to disease progression or death in patients who have a complete or partial response to treatment), overall survival (OS: the time between the first dose of study treatment and death), and progression-free survival (PFS: the time between the first dose of study treatment and death or disease progression, whichever occurs first). Both INV- and IRC-assessed cORR were considered in the comparative efficacy analyses. IRC-assessed DoR was not available for amivantamab at the time of the analyses, so only INV-assessed DoR was included. Only IRC assessment was used for the comparative analyses of PFS in our study because it was used for the analysis of the primary outcomes in both the mobocertinib and amivantamab trials.

Data Sources

Mobocertinib. The mobocertinib phase I/II study (NCT02716116)[10,18] examined the safety, pharmacokinetics, and anti-tumor activity of mobocertinib in patients with locally advanced or metastatic NSCLC. The study included a dose-escalation cohort, an expansion cohort, and an extension cohort. The selection criteria for this analysis included platinum-pretreated patients with *EGFR* ex20ins-positive NSCLC who received mobocertinib 160 mg once daily in the second- or later-line setting in the trial. Median follow-up was 25.8 months, and the data cut-off was November 1, 2021.

Amivantamab. The amivantamab CHRYSALIS phase I trial (NCT02609776) included a dose-escalation cohort and a dose-expansion cohort.[11,19,20] To allow a meaningful comparison with mobocertinib, the present analysis only used data for patients in the dose-expansion cohort: platinum-pretreated adults with advanced NSCLC harboring *EGFR* ex20ins mutations who received amivantamab 1,050 mg (1,400 mg for patients ≥80 kg) once weekly for the first 4 weeks and then once every 2 weeks. Median follow-up was 12.5 months, and the data cut-off was March 30, 2021.

Because individual patient data for OS and IRC-assessed PFS were not available from the amivantamab CHRYSALIS trial, image files of the Kaplan-Meier (KM) curves from CHRYSALIS[19] were converted into coordinates using WebPlotDigitizer version 4.5.[21] These coordinates were then used to reconstruct individual patient data for each curve, as described by Guyot et al.[22] To ensure accuracy, the digitized data were overlaid onto the original image and visually compared. For INV-assessed DoR, data were recreated based on a swimmer plot.[20]

In both studies, patient enrollment was based on local testing to confirm presence of NSCLC *EGFR* ex20ins mutations, and tumor tissue or serum/plasma samples were retrospectively tested to confirm *EGFR* ex20ins mutation variants using next-generation sequencing-based methods.[10,11]

Statistical Analysis

R Software version 3.6.1 was used to perform the statistical analyses.

Matching-Adjusted Indirect Comparison. The MAIC analysis was performed in accordance with guidelines from the UK National Institute for Health and Care Excellence (NICE)[16] and methods described by Signorovitch et al.[14] We first assessed the compatibility of the mobocertinib and amivantamab trials on study design (including eligibility criteria), definition of the outcomes of interest, and availability of similarly defined baseline data for known characteristics that could influence the trial outcomes. These factors were included in the population adjustment (Table 1).

Two scenarios were explored when conducting the population adjustment. In the base case scenario, all commonly available and similarly defined baseline characteristics were included in the population adjustment. The characteristics included age, sex, race, Eastern Cooperative Oncology Group performance status (ECOG PS), histology, smoking history, presence of brain, bone, and liver metastases at baseline, time from diagnosis of advanced NSCLC, number of prior lines of therapy, prior immuno-oncology therapy, and prior EGFR TKI therapy. In the sensitivity analysis scenario, all commonly available and similarly defined prognostic factors and treatment-effect modifiers (as identified through clinical expert consultation) were included in the population adjustment. These characteristics included age, sex, race, ECOG PS, histology, smoking history, number of prior lines of therapy, time from diagnosis of advanced NSCLC, and presence of brain metastases at baseline.

Deriving Balancing Weights. Individual weights for mobocertinib-treated patients were estimated by propensity score-type logistic regression, which leveraged both the patient-level data from the mobocertinib trial and the aggregated data from the amivantamab CHRYSALIS trial. Two sets of weights (one for base case and one for sensitivity analysis) were derived and used to re-weight patients in the mobocertinib trial such that their weighted average characteristics matched those of patients in the amivantamab CHRYSALIS trial. Individual patient weights were also used to calculate the effective sample size (ESS). The ESS and distribution of the weights were evaluated to ensure that the estimated treatment effect was not driven by a small fraction of the patients in the mobocertinib trial. The weights were then applied to the individual patient data from the mobocertinib trial to estimate adjusted efficacy outcomes.

Quantifying the Relative Treatment Effect of Mobocertinib vs. Amivantamab. Relative treatment effects of mobocertinib vs. amivantamab on IRC- and INV-assessed cORR were quantified as odds ratios (ORs) with 95% CIs, before and after population adjustment. The adjusted OR was obtained using a logistic regression analysis fitted on the mobocertinib trial individual patient data and reconstructed individual patient data for the amivantamab CHRYSALIS trial.

For time-to-event outcomes, adjusted KM curves for mobocertinib were estimated by a weighted KM analysis and were used to estimate the median time to an event and survival. The relative effects of mobocertinib vs. amivantamab on IRC-assessed PFS, OS, and INV-assessed DoR were quantified as hazard ratios (HRs) with 95% CIs, before and after population adjustment. The adjusted HRs were obtained using a Cox proportional hazards regression analysis fitted on the mobocertinib trial individual patient data and the reconstructed individual patient data from the amivantamab CHRYSALIS trial. The proportional hazards assumption was tested through visual assessment of Schoenfeld residual plots. The Grambsch-Therneau correlation test was conducted as a global test of proportional hazards.

Robust estimators of variance were used for the derivation of the 95% CIs around the ORs and HRs.

Results

Study Design and Patients in the Mobocertinib and Amivantamab Trials

The mobocertinib and amivantamab trials were comparable in terms of study population, study design, and definitions of the outcomes (Table 1).

Our analysis included 114 patients who received mobocertinib in NCT02716116 and 114 patients who received amivantamab in NCT02609776. Key baseline characteristics are summarized in Table 2**.** Before adjustment, patient characteristics including age, race, sex, and smoking status were notably different between the two trials. After MAIC adjustment, all patient characteristics were balanced between the trials. The ESS was reduced by 33% in the base case model (ESS=76) and by 27% in the sensitivity model (ESS=83) because of the adjustment.

Table 1 Characteristics of the Mobocertinib and Amivantamab Trials

	Mobocertinib NCT02716116	Amivantamab CHRYSALIS NCT02609776
Trial design	Multicenter, open-label, phase I/II	Multicenter, open-label, phase I
Number of patients	114 in the prior-platinum *EGFR* ex20ins pooled cohort	114 in the *EGFR* ex20ins NSCLC efficacy cohort
Median follow-up	25.8 mo	12.5 months
Key inclusion criteria for the populations considered in the analyses	≥18 y Metastatic or unresectable NSCLC ECOG 0-1 *EGFR* ex20ins Prior treatment with platinum therapy	
Treatment administered to the populations considered in the analyses	Mobocertinib 160 mg once daily, oral capsule	Amivantamab 1,050 mg for patients <80 kg (1,400 mg for patients ≥ 80 kg) intravenous once weekly in the first 28-day cycle and once every two weeks in each subsequent cycle
Primary efficacy endpoint	ORR (independent review committee)	ORR (independent review committee)

Abbreviations: ECOG, Eastern Cooperative Oncology Group; *EGFR*, epidermal growth factor receptor; ex20ins; exon 20 insertion mutation; NSCLC, non-small cell lung cancer; ORR, overall response rate.

Table 2 Baseline Demographic and Clinical Characteristics of the Mobocertinib and Amivantamab Populations Before and After MAIC Adjustment in the Base Case Scenario

	Mobocertinib		
Characteristic	**Unweighted (N = 114)**	**Weighted (ESS = 76)**	**Amivantamab (N = 114)**
Age ≥65 y	37%	41%	41%
Race – Asian	60%	52%	52%
Female	66%	61%	61%
Smoking status			
Current/former smoker	29%	43%	43%
Never smoker	71%	57%	57%
ECOG PS			
0	25%	29%	29%
1	75%	71%	71%
Histology			
Adenocarcinoma	98%	96%	96%
Not adenocarcinoma	2%	4%	4%
Presence of metastasis			
Brain	35%	25%	25%
Bone	41%	45%	45%
Liver	21%	11%	11%
Median time from advanced diagnosis to start of therapy[a] (months)	11.8	15.5	15.5
Prior lines of therapy			
One	41%	42%	42%
Two	32%	30%	30%
Three or more	27%	28%	28%
Previous therapy			
Immuno-oncology therapy	43%	44%	44%
EGFR tyrosine kinase inhibitor	25%	20%	20%

Abbreviations: ECOG PS, Eastern Cooperative Oncology Group performance status; EGFR, epidermal growth factor receptor; ESS, effective sample size; MAIC, matching-adjusted indirect comparison.
[a] Mobocertinib or amivantamab.

Table 3 Results of Unanchored MAIC: Base Case Scenario

Outcome	Mobocertinib	Amivantamab[a]	Relative Effect of Mobocertinib vs. Amivantamab	
			Comparison	*P*-Value
Before MAIC				
	Median of KM estimates, months (95% CI)		**HR (95% CI)**	
OS	20.2 (14.9-25.3)	22.8 (17.5-NE)	1.15 (0.76-1.72)	.514
PFS (IRC)	7.3 (5.5-9.2)	6.7 (5.5-9.7)	0.83 (0.60-1.14)	.250
DoR (IRC)	15.8 (7.4-19.4)	10.8 (6.9-15.0)	NE[b]	
DoR (INV)	13.9 (5.6-19.4)	12.5 (6.5-16.1)	0.78 (0.43-1.43)	.426
	Response rate, % (95% CI)		**OR (95% CI)**	
cORR (IRC)	28.1 (20.1-37.3)	43.0 (33.7-52.6)	0.52 (0.30-0.90)	.020
cORR (INV)	35.1 (26.4-44.6)	36.8 (28.0-46.4)	0.93 (0.54-1.60)	.784
After MAIC[c]				
	Median of KM estimates, months (95% CI)		**HR (95% CI)**	
OS	24.8 (18.6-35.5)	22.8 (17.5-NE)	0.92 (0.57-1.48)	.731
PFS (IRC)	7.4 (5.5-12.9)	6.7 (5.5-9.7)	0.74 (0.51-1.07)	.114
DoR (IRC)	16.6 (8.3-19.8)	10.8 (6.9-15.0)	NE[b]	
DoR (INV)	14.2 (7.00-NE)	12.5 (6.5-16.1)	0.59 (0.30-1.18)	.134
	Response rate, % (95% CI)		**OR (95% CI)**	
cORR (IRC)	29.7 (20.0-41.4)	43.0 (33.7-52.6)	0.56 (0.30-1.04)	.067
cORR (INV)	36.5 (25.9-48.4)	36.8 (28.0-46.4)	0.98 (0.53-1.82)	.961

Abbreviations: CI, confidence interval; cORR, confirmed overall response rate; DoR, duration of response; HR, hazard ratio; INV, investigator; IRC, independent review committee; KM, Kaplan-Meier; MAIC, matching-adjusted indirect comparison; NE, not estimable; OR, odds ratio; OS, overall survival; PFS, progression-free survival
[a] The KM estimates for amivantamab were reported in amivantamab European Public Assessment Reports.
[b] Individual patient data for DoR (IRC) were not reported and could not be reconstructed. Thus, DoR (IRC) could not be compared between mobocertinib and amivantamab.
[c] Adjusted for all commonly available and similarly defined baseline characteristics, including age, sex, race, smoking status, Eastern Cooperative Oncology Group performance status, histology, months since diagnosis of advanced non-small cell lung cancer, presence of brain, liver, and bone metastases at baseline, number of prior lines of therapy, prior immuno-oncology therapy, and prior epidermal growth factor receptor (EGFR) tyrosine kinase inhibitor therapy. The effective sample size for the mobocertinib group was 76.

Efficacy Comparisons

Base Case Scenario. IRC-assessed cORR was higher in the amivantamab group than in the mobocertinib group before and after adjustment, with an OR of 0.52 (95% CI: 0.30-0.90) before adjustment and an OR of 0.56 (95% CI: 0.30-1.04) after adjustment (Table 3). INV-assessed cORR was similar between the two groups (OR [95% CI]: 0.93 [0.54-1.60] before adjustment and 0.98 [0.53-1.82] after adjustment).

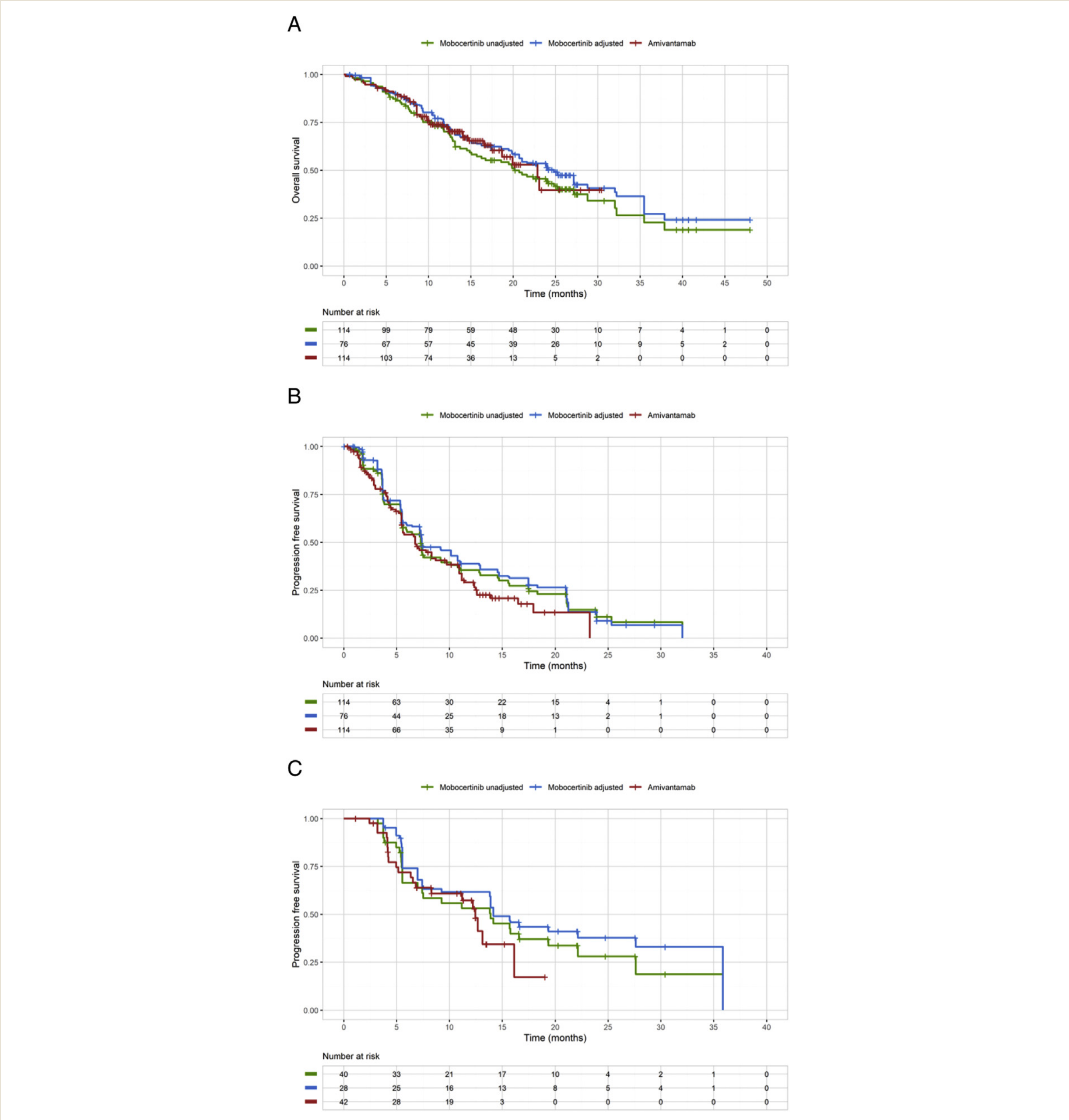
The results of time-to-event outcomes are presented in Table 3, and KM curves are plotted in Figure 1. Treatment with mobocertinib and amivantamab showed similar OS, with an HR of 1.15 (95% CI: 0.76-1.72) before adjustment and 0.92 (95% CI: 0.57-1.48) after adjustment. IRC-assessed PFS was similar between the two groups, with an HR of 0.83 (95% CI: 0.60-1.14) before adjustment and 0.74 (95% CI: 0.51-1.07) after adjustment. Among the responders, INV-assessed DoR was longer in the mobocertinib group compared with the amivantamab group, before and after adjustment: HR 0.78 (95% CI: 0.43-1.43) before adjustment and 0.59 (0.30-1.18) after adjustment. The proportional hazards assumption was valid for the base case scenario (Supplemental Figure 1).

Sensitivity Analysis Scenario. Results were similar between the base case analysis and the sensitivity analysis, which used prognostic factors and treatment-effect modifiers identified by clinical experts (Supplemental Table 1 and Supplemental Figure 2). The proportional hazards assumption was valid for the sensitivity analysis scenario (Supplemental Figure 3).

Discussion

Mobocertinib is an oral *EGFR* TKI[8] and amivantamab is an intravenous monoclonal antibody targeting *EGFR* and c-Met.[9] Both recently received approval in the US for the treatment of locally advanced or metastatic *EGFR* ex20ins+ NSCLC that has progressed during or after platinum-based chemotherapy. However, their efficacy has not been compared in head-to-head clinical trials. We compared their efficacy using an unanchored MAIC accounting for cross-study differences in all commonly available and similarly defined patient baseline characteristics for the base case scenario and in baseline characteristics considered to be prognostic factors or treatment-effect modifiers in the sensitivity analysis scenario. We found that mobocertinib and amivantamab were equivalent in terms of INV-assessed cORR, IRC-assessed PFS, OS, and INV-assessed DoR before and after adjustment. However, IRC-assessed cORR was considerably higher in the amivantamab group than in the mobocertinib group (before adjustment), while INV-assessed cORR was similar between the two groups. In addition, INV-assessed cORR was higher than IRC-assessed cORR in the mobocertinib trial, while the opposite was observed in the amivantamab CHRYSALIS trial.[10,11] Variability between IRC and INV assessments is not uncommon; therefore, IRC and INV comparisons should be interpreted with caution. Additionally, in both trials, IRC- and INV-assessed PFS were similar in patients treated with amivan-

Figure 1 Kaplan-Meier curves of time-to-event outcomes for mobocertinib vs. amivantamab in the base case scenario. Kaplan-Meier plots are shown for overall survival (A), progression-free survival assessed by an independent review committee (B), and duration of response assessed by investigator review (C).



tamab or mobocertinib. As a result, it was expected that using MAIC to compare the relative efficacy of INV-assessed PFS for mobocertinib vs. amivantamab would have yielded similar results to IRC-assessed PFS.

Recently, Van Sanden et al. conducted an unanchored MAIC study to estimate the relative effect of amivantamab vs. mobocertinib in platinum-pretreated patients with *EGFR* ex20ins-mutated NSCLC by comparing individual patient data from the amivantamab CHRYSALIS trial against published data from the mobocertinib phase I/II trial.[23] Although their study adjusted for fewer variables than our study, their results were consistent with our findings. Their study matched for the number of prior lines of therapy, ECOG PS, presence of brain metastasis, age, race, sex, smoking history, and histology but not prior immunotherapy or prior EGFR/human epidermal growth factor receptor TKI therapy. The results of that study showed that amivantamab had a higher IRC-assessed cORR than mobocertinib, but both treatments showed similar INV-assessed cORR, IRC- and INV-assessed clinical benefit rates, IRC-assessed PFS, and OS.

A strength of the current study and the Van Sanden et al. study[23] is that the MAIC method enabled relative efficacy data to be analyzed in the absence of head-to-head evidence. By balancing the populations on key baseline patient characteristics, the MAIC reduced the bias for measuring efficacy between the two treatments compared to that of an unadjusted indirect comparison.

Our study had several limitations. The potential for residual confounding exists due to unavailable or unreported patient characteristics in the included studies. In particular, prior surgery for NSCLC, prior chemo- and radiotherapy for stage II or III NSCLC, and cancer stage at enrollment (stage IIIB vs. IV) could not be included in the population adjustment while being mentioned as potential prognostic factors or treatment-effect modifiers in advanced NSCLC. Unobserved imbalances in these characteristics may have influenced the treatment outcomes and thus confounded the estimates of treatment effect measured on the matched populations. Additionally, the absence of patient-level individual data from the amivantamab CRYSALIS trial warrants caution in the interpretation of the results.

Some uncertainties were also noted in terms of outcome definitions and measurements across studies. Precise definitions of the censoring rules used in the analysis of PFS and DoR were not reported in the amivantamab CHRYSALIS trial. Therefore, differences in the censoring rules used in the mobocertinib and amivantamab trials may exist, which could have biased the estimation of relative efficacy of mobocertinib vs. amivantamab. Moreover, the two studies had different timings for disease assessments. Specifically, the mobocertinib trial had 8-weekly disease assessment visits in the first 14 cycles followed by a monitoring visit every 12 weeks thereafter, while the amivantamab CHRYSALIS trial had 6-weekly monitoring visits. This difference may have caused assessment-time bias in favor of mobocertinib if initial progressive disease events were captured later for patients treated with mobocertinib due to later disease assessment. Additionally, both trials enrolled patients based on local testing, and not all patients had their *EGFR* ex20ins variants confirmed by central sequencing. Data about patients with confirmed *EGFR* ex20ins in the amivantamab group were not available. Therefore, we could not conduct subgroup analyses on this data. Lastly, after adjustment, the ESSs were reduced by 33% in the base case model and 27% in the sensitivity model. This reduction in sample size may have reduced the ability to detect differences in treatment effects.

Conclusions

The present analysis suggested similar efficacy of mobocertinib and amivantamab in patients with NSCLC harboring *EGFR* ex20ins mutations that has progressed during or after platinum-based chemotherapy. Having multiple treatment options with distinct mechanisms of action and safety profiles may benefit the needs of different patients. Indeed, where efficacy is comparable, differences in toxicities and other treatment attributes such as route of delivery and dosing schedule may help determine which drug is most appropriate for a given patient. After this study was conducted, the sponsor (Takeda) decided to withdraw mobocertinib. Nevertheless, the MAIC methodology and study results provide valuable information for physicians choosing from different treatment options in the future.

Clinical Practice Points

- Efficacy against *EGFR* ex20ins+ non-small cell lung cancer (NSCLC) has been demonstrated for mobocertinib (an epidermal growth factor receptor [EGFR] tyrosine kinase inhibitor [TKI]) and amivantamab (which targets both EGFR and c-Met). However, these drugs have not been compared in a head-to-head trial.
- To address this knowledge gap, we conducted a matching-adjusted indirect comparison using data from a phase I/II single-arm trial of mobocertinib (NCT02716116) and a phase I single-arm trial of amivantamab (NCT02609776).
- Trial participants (N=114 for each trial) had experienced disease progression during or after platinum-based chemotherapy.
- Confirmed overall response rate, progression-free survival, overall survival, and duration of response were assessed.
- Clinical efficacy outcomes were generally comparable for mobocertinib and amivantamab.
- These findings confirm the efficacy of treatment options for *EGFR* ex20ins+ NSCLC with different mechanisms of action, which may help fulfil the different needs of patients.

CRediT authorship contribution statement

Sai-Hong Ignatius Ou: Conceptualization, Visualization, Writing – review & editing. **Thibaud Prawitz:** Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – review & editing, Supervision, Project administration. **Huamao M. Lin:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision, Project administration. **Jin-liern Hong:** Conceptualization, Methodology, Writing – review & editing. **Min Tan:** Methodology, Software, Validation, Formal analysis, Writing – review & editing. **Irina Proskorovsky:** Conceptualization, Methodology, Writing – review & editing, Supervision, Project administration. **Luis Hernandez:** Conceptualization, Methodology, Validation, Writing – review & editing. **Shu Jin:** Data curation, Supervision. **Pingkuan Zhang:** Conceptualization, Writing – review & editing, Supervision. **Jianchang Lin:** Methodology, Formal analysis, Writing – review & editing. **Jyoti Patel:** Conceptualization, Methodology, Resources, Writing – review & editing. **Danny Nguyen:** Writing – review & editing. **Joel W. Neal:** Resources, Writing – review & editing, Supervision.

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Disclosures

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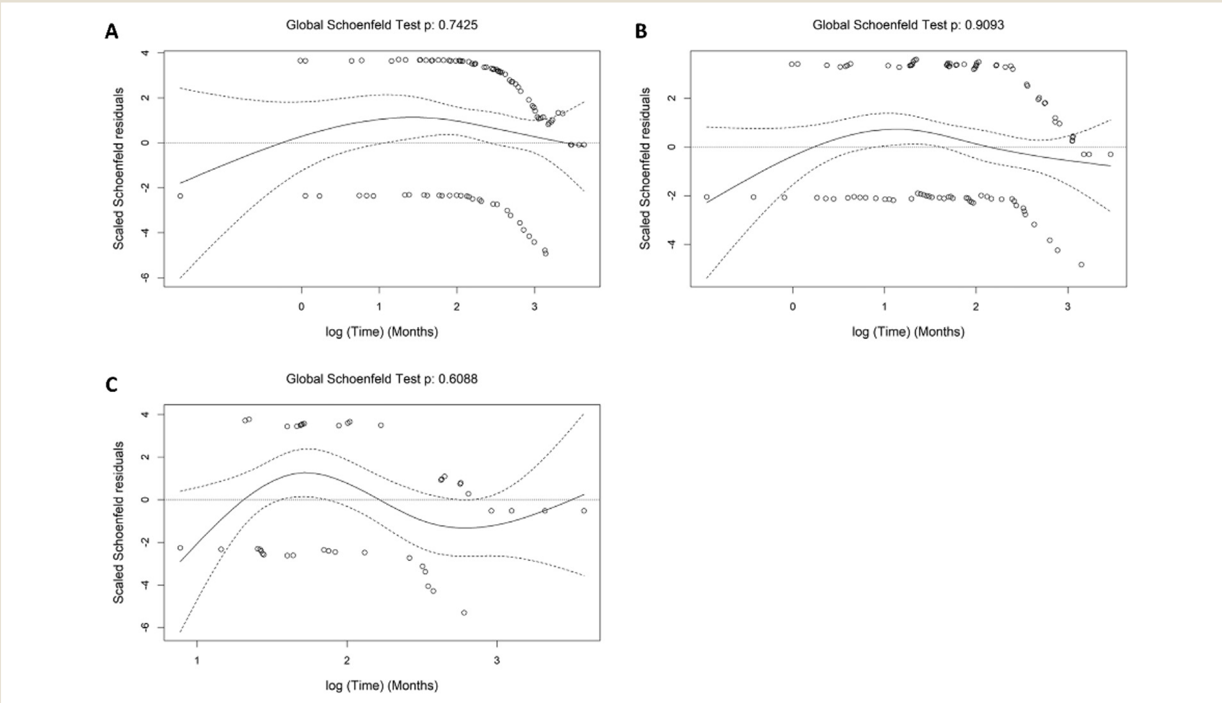
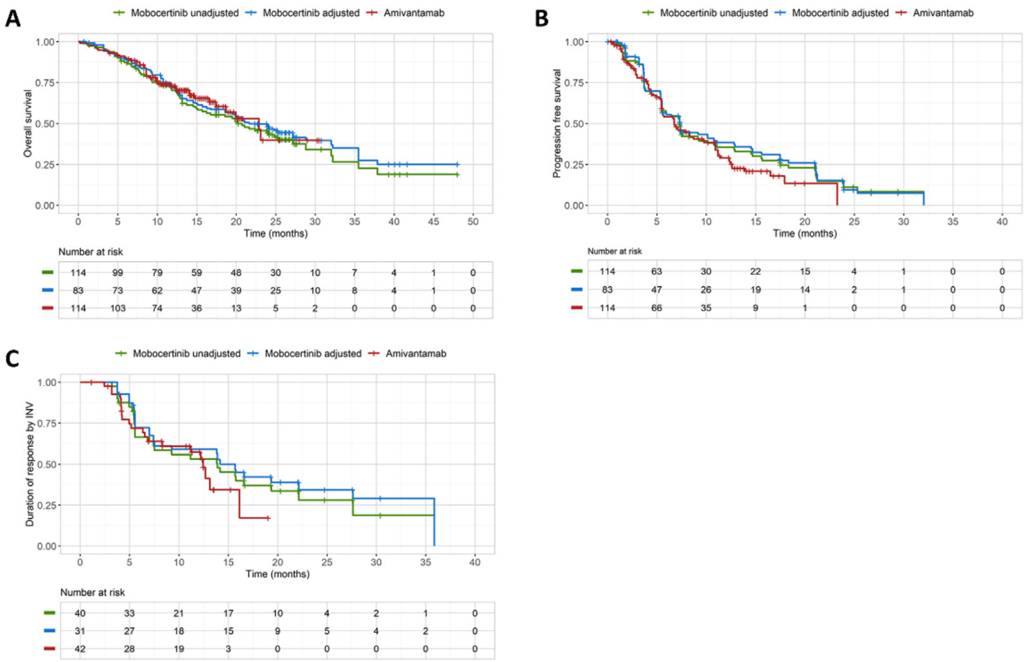
Supplementary materials

Supplemental Figure 1 Assessment of the proportional hazards assumption in the base case scenario. P-values were calculated with the Grambsch and Therneau test. Schoenfeld plots are shown for overall survival (A), progression-free survival assessed by an independent review committee (B), and duration of response assessed by investigator review (C).



Supplemental Figure 2 Kaplan-Meier curves of time-to-event outcomes for mobocertinib vs. amivantamab in the sensitivity analysis scenario. Kaplan-Meier plots are shown for overall survival (A), progression-free survival assessed by an independent review committee (B), and duration of response assessed by investigator review (C).



Supplemental Table 1 Results of Unanchored MAIC: Sensitivity Analysis Based on Factors Identified by Clinical Experts

Outcome	Mobocertinib	Amivantamab[a]	Relative Effect of Mobocertinib vs. Amivantamab	
			Comparison	*P*-value
Before MAIC				
	Median of KM estimates, months (95% CI)		**HR (95% CI)**	
OS	20.2 (14.9-25.3)	22.8 (17.5-NE)	1.15 (0.76-1.72)	.514
PFS (IRC)	7.3 (5.5-9.2)	6.7 (5.5-9.7)	0.83 (0.60-1.14)	.250
DoR (IRC)	15.8 (7.4-19.4)	10.8 (6.9-15.0)	NE[b]	
DoR (INV)	13.9 (5.6-19.4)	12.5 (6.5-16.1)	0.78 (0.43-1.43)	.426
	Response rate, % (95% CI)		**OR (95% CI)**	
cORR (IRC)	28.1 (20.1-37.3)	43.0 (33.7-52.6)	0.52 (0.30-0.90)	.020
cORR (INV)	35.1 (26.4-44.6)	36.8 (28.0-46.4)	0.93 (0.54-1.60)	.784
After MAIC[c]				
	Median of KM estimates, months (95% CI)		**HR (95% CI)**	
OS	21.7 (15.1-32.2)	22.8 (17.5-NE)	1.01 (0.64-1.60)	.950
PFS (IRC)	7.4 (5.5-12.8)	6.7 (5.5-9.7)	0.77 (0.54-1.11)	.161
DoR (IRC)	16.6 (6.5-20.3)	10.8 (6.9-15.0)	NE[b]	
DoR (INV)	14.2 (7.00-NE)	12.5 (6.5-16.1)	0.64 (0.33-1.26)	.198
	Response rate, % (95% CI)		**OR (95% CI)**	
cORR (IRC)	31.3 (21.8-42.6)	43.0 (33.7-52.6)	0.60 (0.33-1.11)	.102
cORR (INV)	37.8 (27.6-49.2)	36.8 (28.0-46.4)	1.04 (0.58-1.89)	.888

Abbreviations: CI, confidence interval; cORR, confirmed overall response rate; DoR, duration of response; HR, hazard ratio; INV, investigator; IRC, independent review committee; KM, Kaplan-Meier; MAIC, matching-adjusted indirect comparison; NE, not estimable; OR, odds ratio; OS, overall survival; PFS, progression-free survival.
[a] The KM estimates for amivantamab were reported in amivantamab European Public Assessment Reports.
[b] Individual patient data for DoR (IRC) were not reported and could not be reconstructed. Thus, DoR (IRC) could not be compared between mobocertinib and amivantamab.
[c] Adjusted for factors identified by clinical experts, including age, sex, race, smoking status, Eastern Cooperative Oncology Group performance status, histology, months since diagnosis of advanced non-small cell lung cancer, presence of brain metastases at baseline, and number of prior lines of therapy. The effective sample size for the mobocertinib group was 83.

Supplemental Figure 3 **Assessment of the proportional hazards assumption in the sensitivity analysis. Schoenfeld residual plots are shown for overall survival (A), progression-free survival assessed by an independent review committee (B), and duration of response assessed by investigator review (C).**